Accepted Manuscript

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PII: S0140-7007(13)00259-4

DOI: 10.1016/j.ijrefrig.2013.09.025

Reference: JIJR 2632

To appear in: *International Journal of Refrigeration*

Received Date: 30 April 2013

Revised Date: 3 September 2013

Accepted Date: 15 September 2013

THE PERFORMANCE OF A LARGE-SCALE ROTARY MAGNETIC REFRIGERATOR

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ABSTRACT

Astronautics has constructed a large-scale rotary magnetic refrigerator which was designed to provide 2 kW of cooling power over a temperature span of 12 K with electrical Coefficient of Performance (COPe) > 2. The system uses a NdFeB magnet assembly with peak field of 1.44 tesla which rotates over twelve beds arranged circumferentially. Each bed was packed with six layers of LaFeSiH of different Curie temperatures, chosen to optimize system performance over the desired span. We report here on the performance of this system at flow rates ranging from 12.5 to 21.2 liter min^{-1}. At the largest flow rate, the system produced 3042 W of cooling power at zero span and peak performance of 2502 W over a span of 11 K. To our knowledge, this represents the largest cooling power yet observed for a magnetic refrigeration system. We show that the measured performance is in good agreement with theoretical prediction.

Keywords: magnetic refrigeration, magnetocaloric material, active magnetic regeneration, COP

Nomenclature

Acronyms

AMR	Active Magnetic Regenerator
COPe	Electrical coefficient of performance
DSC	Differential scanning calorimetry
MCM	Magnetocaloric material
MR	Magnetic refrigeration

Symbols

A	Bed cross-sectional area (cm^2)
a	Linear coefficient for temperature-dependent fluid viscosity (Pa s)
b	Quadratic coefficient for temperature-dependent fluid viscosity (Pa s)
B_0	Peak magnetic flux density (tesla)
C_e	Heat capacity of epoxy (J $kg^{-1}K^{-1}$)
C_{es}	Effective solid-phase heat capacity (J $kg^{-1}K^{-1}$)
C_f	Fluid heat capacity (J $kg^{-1}K^{-1}$)
C_{MCM}	Heat capacity of the magnetocaloric material
D_p	Particle diameter (μm)

h	Fluid-particle surface heat transfer coefficient (W m^{-2} K^{-1})
k_f	Static fluid thermal conductivity (W m^{-1} K^{-1})
k_{ef}	Effective fluid-phase thermal conductivity (W m^{-1} K^{-1})
Pr	Prandtl number
Q_C	Cooling power (W)
Q_H	Heat exhaust (W)
r	Epoxy:magnetocaloric material mass ratio
Re	Reynolds number
V_B	Total bed volume (liter)
T	Temperature (K)
T_{Ci}	Cold inlet fluid temperature (°C)
T_{Co}	Cold outlet fluid temperature (°C)
T_{Hi}	Hot inlet fluid temperature (°C)
T_{Ho}	Hot outlet fluid temperature (°C)
ε	Bed porosity
μ	Specific exergetic cooling power (W $tesla^{-1}$ $liter^{-1}$)
μ_0	Reference fluid viscosity (Pa s)
μ_f	Fluid viscosity (Pa s)
ω	Angular velocity of magnet assembly (degrees s^{-1})
ρ_f	Fluid density (kg m^{-3})
Φ	System flow rate (liter min^{-1})
Φ_b	Bed flow rate (liter min^{-1})

1. INTRODUCTION

With the development of powerful NdFeB magnets and the discovery of materials with a so-called giant magnetocaloric effect, room temperature magnetic refrigeration (MR) has the potential to supplant vapor compression for many typical cooling applications (Gschneidner and Pecharsky, 2008). Magnetic cooling systems, in theory, provide a significant increase in energy efficiency relative to vapor compression systems (Engelbrecht *et al.*, 2007), which would lead to reduced operating costs and energy consumption. In addition, unlike vapor compression, MR uses no greenhouse or ozone-depleting gases, making it a truly green technology. Finally, the (typically) aqueous heat transfer fluid used in an MR system is much less likely to leak than the gases used in vapor compression, and the leaks can be readily identified and corrected if they should occur. Modern room-temperature MR systems implement the so-called Active Magnetic Regenerator (AMR) cycle to perform cooling (Barclay and Steyert, 1982). The AMR used in this cycle consists of a porous bed of magnetocaloric material (MCM). A heat transfer fluid exchanges heat with the MCM as it flows through the bed. The cycle has four stages. In the first stage ("magnetization"), while the fluid in the bed is stagnant, a magnetic field is applied to the MCM, causing it to heat. In the next stage (the "hot blow"), while the magnetic field over the bed is maintained, fluid at a temperature T_{Ci} (the cold inlet temperature) is pumped through the bed from the cold side to the hot side. This fluid pulls heat from the MCM and rises in temperature as it passes through the bed. During the hot blow, the fluid exits the bed at the temperature T_{Ho} (the hot outlet temperature) and is circulated through a heat exchanger, where it gives up heat to the ambient environment and returns to the temperature T_{Hi} (the hot inlet temperature) < T_{Ho}. In the next stage ("demagnetization"), the fluid flow is terminated and the magnetic field is removed. This causes the bed to cool further. In the final stage (the "cold blow"), fluid at a temperature T_{Hi} is pumped through the bed from the hot side to the cold side in the continued absence of the magnetic field. The fluid is cooled as it passes through the MCM, reaching a temperature T_{Co} (the cold outlet temperature) ≤ T_{Ci}. The colder fluid exiting the bed during the cold blow may be circulated through a cold-side heat exchanger, picking up heat from a refrigerated environment, allowing it to maintain its colder temperature. The fluid exits the cold-side heat exchanger at temperature T_{Ci} and completes the AMR cycle. A number of prototype magnetic refrigeration systems, all implementing the AMR cycle, have been reported in the literature (see, for example, Engelbrecht *et al*., 2012, Tura and Rowe, 2011, Zimm *et al*., 2007) and demonstrate that the magnetic refrigeration process is capable of providing cooling power (50 W – 1000 W) comparable to that delivered by common household appliances. However, to justify the significant investment that will be required to bring this technology to the marketplace, ever higher performance in cooling power, span, and efficiency must be demonstrated, with systems of reasonable size. To this end, Astronautics Corporation of America has constructed a large-scale MR system designed to provide at least 2 kW of cooling power over a temperature span of 12 K with COP of at least 2. The system, constructed with funding provided by the US Office of Naval Research, is intended to meet the preliminary performance specifications of a supplemental electronics cooler, a compact ship-board cooling system that provides additional localized cooling to critical electronics on a naval vessel during operation in warm-water environments. The system is intended for operation above room temperature, with a hot inlet fluid temperature of 44 °C and cold inlet fluid temperature of 32 °C.

2. SYSTEM DESCRIPTION

The Astronautics MR system uses a rotating magnet architecture where a magnet assembly producing a peak field of 1.44 tesla rotates over twelve fixed beds arranged circumferentially. Each bed is separated by an angle of 360°/12 = 30° from its neighbors. The magnet assembly, in the form of a modified Halbach array (Chell, 2008), was designed by Astronautics and fabricated by Vacuumschmelze GmbH & Co. KG. Beds in the system fit into a gap in this assembly. Each bed has four ports: the hot inlet, hot outlet, cold inlet, and cold outlet. Flow through the beds is controlled by four rotary disk valves referred to as the hot inlet valve, cold outlet valve, cold inlet valve, and hot outlet valve, which are mechanically coupled to the rotation of the magnet. Each bed in the system undergoes the same AMR cycle, with a hot blow duration of 1/3 of a cycle, a demagnetization stage with no flow lasting 1/6 of a cycle, a cold blow duration of 1/3 of a cycle, and a magnetization stage with no flow lasting 1/6 of a cycle. There is a time delay of 30°/ω between the cycles experienced by adjacent beds, where ω is the angular rotation rate of the magnet (in degrees s^{-1}). The system architecture arranges beds in pairs, with beds in the pair 180° out of phase in their refrigeration cycles: while one bed in the pair is magnetized (inside the magnet gap) and undergoes its hot blow stage, the second bed in the pair is demagnetized (outside of the magnet gap) and undergoes its cold blow stage. Figure 1 gives a schematic view of flow in the system, focusing on one pair of beds. When the magnet rotates over a bed (shown on the left), the cold inlet valve port connected to the cold inlet bed port opens, delivering flow from the valve to the cold inlet bed port. Simultaneously, the hot outlet valve port connected to the hot outlet bed port also opens, collecting flow exiting the hot outlet bed port. Flow is therefore directed from the cold inlet of the bed to the hot outlet, implementing the hot blow of the AMR cycle for the magnetized bed. At the same time, the hot inlet valve directs flow to the hot inlet bed port of the demagnetized bed in the pair (shown on the right), while the cold outlet valve port simultaneously opens to collect flow coming from the cold outlet bed port, implementing the cold blow of the AMR cycle. The valves block all other bed ports in this pair so that flow proceeds through the system along the solid black line shown in Figure 1. As the magnet rotates away from the bed on the left and over the bed on the right, the valves first shut off all flow in the beds by blocking all the bed ports, and then switch the flow into the alternate path shown as the dashed line in the figure. This flow proceeds in the opposite direction through the bed pair. We note that with this architecture, flow in all pipes (solid and dashed lines in the figure) is uni-directional and there is no mixing of inlet and outlet fluid. Volume where inlet and outlet flow can mix is referred to as “dead volume”, and its presence degrades system performance (Jacobs, 2009). At any instant in time, four beds are inside the magnet gap and are undergoing their hot blow, their four partners are outside of the magnet gap and are undergoing their cold blow, and four beds (2 pairs) are in transition between being magnetized/demagnetized and have no flow. The collected flow from all the cold outlet ports exits the cold outlet valve and could be routed through a cold-side heat exchanger. However, in the current configuration of the system, the cold outlet fluid is routed through an electrical heater. This heater provides the cooling load for the system; by controlling the heater current and voltage drop, the cooling load can be easily and accurately controlled and recorded. The collected flow from the hot outlet ports is returned to the pump and is then sent through a hot-side heat exchanger for heat exhaust before proceeding to the hot inlet valve. The other side of the hot-side heat exchanger is connected to a temperature-controlled bath. The bath temperature is adjusted so that the temperature of the fluid exiting the heat exchanger (T_{Hi}) has a desired value. While the system reported here is completely new, it uses the same rotary magnet architecture and valving that were used in an earlier MR prototype; for further details the reader may consult Russek *et al*., 2010, Zimm, *et al*., 2007, and Zimm, *et al*., 2003.

Each bed in this system has a length of 3.8 cm and a cross-sectional area of 7.9 cm^2. Consequently, it can house 30 cm^3 of magnetocaloric material. Each bed is packed with six layers of $La(FeSi)_{13}H$ (Fujita *et al*., 2003) with Curie temperatures chosen to optimize theoretical performance over a span of 12 K with T_{Hi} = 44 °C. The Curie temperatures of the bed layers are given in Table 1, where the Curie temperature is taken to be the temperature of the peak in the zero-field heat capacity. The measured zero-field heat capacities of the six layers used in the system, obtained using a DSC (heating), are shown in Figure 2. The peak entropy change and adiabatic temperature change of this material in a field of 1.5 tesla, evaluated from measurement of field-dependent heat capacity (Mudryk and Pecharsky, 2010) are typically 14 J kg^{-1}K^{-1} and 4 K, respectively. The material exhibits small thermal hysteresis, with a temperature difference of approximately 2.5 K between the peak zero-field heat capacities under heating and cooling. However, we have not corrected this difference to account for instrumental hysteresis. The material also exhibits small magnetic hysteresis of about 0.07 tesla.

The LaFeSiH used in the beds had a measured density of 6.7 g cm^{-3} and was in the form of spherical particles with diameters between 177 and 246 μm. The system used a total LaFeSiH mass of 1.52 kg. The beds were fabricated with a modified version of the epoxy-connection process described by Barclay and Merida-Donis (1997). The epoxy-connected layers were tested for durability prior to use in the system and were shown to maintain their integrity under combined field and flow cycling. The porosity was measured on two beds by direct evaluation of the pore volume and was found to be 36.7% and 37.5%. An average porosity of 37% was then assumed for all beds in the system. The pressure drop as a function of flow rate was measured for each bed and the averaged results are presented in Figure 3. The error bars indicate ± one standard deviation. There was little variation in pressure drop among the beds. The figure also shows the pressure drop expected from the Ergun-MacDonald equation (Kaviany, 1995) with particle diameters at the lower and upper bounds of the particles used in the beds. The measured pressure drop favored the lower end of the expected range, which may have resulted from reduced drag on the smooth epoxy coating on the particles.

It is known that LaFeSiH experiences degradation in its magnetocaloric properties when held close to its Curie temperature (Barcza *et al*., 2011). The authors have found, however, that this effect is completely reversible by holding the material sufficiently far away from its Curie temperature (Zimm and Jacobs, 2013, Russek *et al*., 2011). For example, Figure 4 (left panel) shows a DSC trace of a sample of LaFeSiH that had been held near its Curie temperature for over one year. The DSC trace, which initially had been a single sharp peak centered at 28 °C, has degraded into a broad feature with two shallow peaks. The right panel of Figure 4 shows the DSC trace of the same material after being held at 60 °C for 72 hours. The high temperature exposure has completely reversed the degradation of the material. While recovery proceeds more quickly at elevated temperatures, it will also occur when the degraded material is held sufficiently far below its Curie temperature, and we note from Table 1 that at room temperature (23 °C), all layers will be at least 7 K below their Curie temperature. Thus, when the magnetic refrigerator is idle at room temperature, which occurs at least every night and over weekends, any peak splitting degradation that may have occurred during operation is completely reversed. As a result, the performance of the system has remained stable over 12 months of operation.

A photograph of the system is shown in the left panel of Figure 5. After initial validation testing and debugging, the system was extensively insulated, as shown in the right panel of Figure 5. We note that the system is intended to operate above room temperature so the insulation was necessary to prevent heat leaks to ambient, which could have artificially increased the apparent cooling power. In addition to the 5 cm-thick insulating panels surrounding the system (seen in the right of Figure 5), individual tubes carrying flow to the beds were placed in insulating sleeves, and internal insulation was placed around the cold side of the system. The insulated cube housing the system measures 1.02 m × 0.81 m × 0.91 m.

To establish the efficacy of the insulation, fluid initially at a temperature of 44 °C was pumped through the system at a flow rate of 10 liter min^{-1}. To prevent any magnetic refrigeration effects, the magnet was kept stationary during this test. The system was then allowed to reach a steady-state. After traversing through the insulated system, the fluid temperature returning to the pump was 43.9 °C, 0.1 °C below its starting value, corresponding to a heat leak of 77 W. In contrast, in the absence of insulation, the heat leak measured in the same manner was 431 W. The accuracy of the thermistors used to measure the fluid temperatures, as stated by the manufacturer, is 0.05 °C. The accuracy in the difference between two thermistor readings is therefore expected to be approximately $\sqrt{2}\times 0.05 \approx 0.07$ °C, which is comparable to the measured difference between the initial fluid temperature and its value after passing through the insulated system. We also note that during operation as a refrigerator, as the temperature span increases the cold side of the system becomes closer to room temperature, further decreasing heat leaks to ambient.

3. SYSTEM PERFORMANCE

We show in Figure 6 the measured cooling power of the system as a function of temperature span at flow rates of 12.5, 15.5, 19.4, and 21.2 liter min^{-1}. The refrigeration cycle frequency in all cases was 4

Hz; at present, we have not attempted to optimize the cycle frequency for the different flow rates. The cooling power is defined as

$$Q_C = \rho_f C_f \Phi (T_{Ci} - T_{Co}) \tag{1}$$

where ρ_f = 1000 kg m^{-3} and C_f = 4180 J kg^{-1} K^{-1} are the fluid density and heat capacity, respectively, Φ is the system flow rate, and T_{Ci} and T_{Co} are the measured cold inlet and cold outlet fluid temperatures. The temperature span is defined to be $T_{Hi} - T_{Ci}$, where T_{Hi} is the hot inlet fluid temperature. The fluid temperatures are obtained from individually calibrated thermistors installed in the fluid lines; as noted above, thermistor accuracy is 0.05 °C, while flow meter accuracy, based on comparison with timed volume collection, is approximately 0.5 liter min^{-1}. Water, with a slight amount of an anti-corrosion agent, was used as the heat transfer fluid in the system. Here and in the following, the hot inlet fluid temperature was fixed at T_{Hi} = 44 °C; the span was obtained by reduction of T_{Ci}.

The Astronautics system has produced peak zero-span cooling power of 3042 W, and 2090 W of cooling power over a span of 12 K, slightly exceeding its design target. To the best of our knowledge, the system has achieved the highest cooling powers yet reported for a magnetic refrigeration system.

As noted above, the cooling load in the system is provided by an electrical resistance heater. The heater power consumption is monitored during system operation and can be used to provide an independent check of the cooling power given by (1). Figure 7 compares the heater power consumption and the cooling power as a function of the cooling power over a number of different trials. There is generally excellent agreement between the two, although the heater power consumption is slightly smaller. This is due, in part, to the additional fluid power gain associated with overcoming the pressure drop of the heater, which is not included in the heater power consumption. The solid line in the figure represents equality and is included as a guide to the eye.

In Figure 8, we show the specific exergetic cooling power (Rowe, 2011), defined as

$$\mu = Q_C \left(\frac{T_{Hi} - T_{Ci}}{T_{Ci}} \right) \frac{1}{B_0 V_B} \tag{2}$$

where B_0 is the peak magnetic field and V_B is the total bed volume. The best performance of the system to date, using (2) as the metric, was obtained at 21.2 lit/min with 2502 W of cooling power delivered over a temperature span of 11.0 °C, corresponding to μ = 173 W tesla^{-1} liter^{-1}.

The power consumption of the drive motor (which rotated the magnet assembly and valves) and the pump were monitored during system operation. We define the power consumption of the refrigeration system to be the sum of the power consumption of these two components. We then define the electrical Coefficient of Performance (COPe) to be the ratio between the cooling power delivered by the system to the power consumption. This ratio is shown in Figure 9 for system operation at 15.5, 19.4, and 21.2 liter min^{-1}. Somewhat surprisingly, the COPe actually increased slightly as the flow rate was raised from 19.4 to 21.2 liter min^{-1}. This increase resulted from an increase in pump efficiency at the higher flow rate.

For temperature spans below about 10 K, the COPe of the system remained above 2. At its point of peak performance (as measured by the specific exergetic cooling power), the system operated with COPe = 1.9. At its design operating point (2 kW over a 12 K span), the COPe was 1.6, somewhat below the target of 2. The COPe depends on the efficiency of the drive motor and pump, and these efficiencies were mediocre. At the highest flow rate, the drive motor efficiency averaged 70%, while the pump efficiency averaged 35%. High-efficiency electrical motors can operate at almost 90% efficiency and we had obtained pump efficiency of 43% on a smaller magnetic refrigeration prototype. The COPe of the system could be significantly improved by using components with higher efficiencies.

3. COMPARISON WITH THEORETICAL PERFORMANCE

To model the theoretical performance of the magnetic refrigeration system, we use the solid and fluid phase temperature profile equations derived by Engelbrecht (2008). We impose Danckwerts boundary conditions (Wakao and Kaguei, 1982) at the axial ends of the bed for the fluid phase temperature and insulating boundary conditions for the solid phase. To incorporate the effect of the epoxy used to form inter-particle connections in the beds, we employ a mass-weighted average of the thermal, physical, and magnetocaloric properties of the magnetocaloric material and the epoxy. For example, the effective heat capacity of the solid phase is taken to be

$$C_{es} = \frac{C_{MCM} + rC_e}{1+r} \tag{5}$$

where C_{MCM} is the heat capacity of the magnetocaloric material, C_e is the heat capacity of the epoxy, and r is the ratio of the epoxy mass to magnetocaloric material mass in the bed. The magnetocaloric properties of the magnetocaloric material used in the modeling were based on temperature and field-dependent heat capacity data provided by Ames Laboratory (Mudryk and Pecharsky, 2010). We use the surface heat transfer coefficient between the solid and fluid phases suggested by Wakao and Kaguei (1982):

$$h = \frac{k_f}{D_p}\left(2 + 1.1\mathrm{Pr}^{1/3}\,\mathrm{Re}^{0.6}\right) \tag{6}$$

where

$$\mathrm{Pr} = \frac{C_f \mu_f}{k_f} \tag{7}$$

and

$$\mathrm{Re} = \frac{\rho_f \Phi_b D_p}{A\mu_f} \tag{8}$$

are the Prandtl and Reynolds numbers, respectively, k_f = 0.6 W m^{-1}K^{-1} is the static fluid thermal conductivity, D_p = 212 μm and A are the average particle diameter and bed cross-sectional area, respectively, and Φ_b is the bed flow rate during a blow. This bed flow rate is always one fourth of the system flow rate Φ. We use a temperature-dependent fluid viscosity given by

$$\mu_f(T) = \mu_0 + ax + bx^2 \tag{9}$$

with $x = (T - T_0)/(T_1 - T_0)$, T_0 = 275 K, T_1 = 300 K, μ_0 = 1.65×10^{-3} Pa s, a = -1.03×10^{-3} Pa-s, and b = 2.35×10^{-3} Pa s. Finally, we use an effective fluid-phase thermal conductivity that includes thermal dispersion. Specifically, we use the form suggested by Wakao and Kaguei (1982) :

$$k_{ef} = k_f(\varepsilon + 0.5\,\mathrm{Pr}\times\mathrm{Re}) \tag{10}$$

where ε = 0.37 is the bed porosity.

Each bed in the Astronautics system has a small chamber at each end where fluid emerging from an inlet pipe can homogenize spatially before entering the magnetically active portion of the bed. Within these chambers, inlet fluid about to enter the bed can mix with outlet fluid remaining in the chamber from a previous blow; that is, these chambers represent dead volume in the system. We incorporate the effect of these chambers in the model by treating them as layers with 100% porosity.

By numerical solution of the Engelbrecht model equations, we obtain the solid and fluid phase temperature profiles. A discussion of the numerical solution process is outside the scope of this paper; here, we note only that it results in temperature profiles that are time-periodic, continuous across the boundaries between bed layers, satisfy the Engelbrecht model equations with very small error, and satisfy the boundary conditions. From knowledge of the fluid temperature profile in the bed, we can evaluate its predicted cooling power and heat exhaust. Because each bed in the Astronautics system undergoes an identical refrigeration cycle, the predicted cooling power and heat exhaust of the system will be the product of the single-bed results obtained above with the number of beds (12) in the system. We show in Figure 10 the measured and predicted cooling power of the Astronautics system as functions of temperature span for flow rates of 12.5, 15.5, and 19.4 liter min^{-1}. Below a temperature span of 16 K, good agreement between measurement and prediction is obtained at all flow rates, although at the lowest flow rate the system slightly underperforms relative to theory, and slightly overperforms at the highest flow rate. Above 16 K, the theoretical model predicts longer tails to the cooling power that are not observed in actual system operation. While the source of this discrepancy is not known, it may result from loss mechanisms (e.g., internal heat leaks, eddy currents, magnetic or thermal hysteresis) that are not included in the model. Alternatively, the tail could be an unphysical artifact of the model or of the magnetocaloric properties used in the model. Its cause is under investigation.

In Figure 11, we show the measured and predicted heat exhaust at the same flow rates as Figure 10, where the measured heat exhaust is given by

$$Q_H = \rho_f C_f \Phi\left(T_{Ho} - T_{Hi}\right) \tag{6}$$

and T_{Ho} is the hot outlet fluid temperature. Again, generally good agreement between theory and measurement is obtained, although the discrepancy in heat exhaust at the highest flow rate has increased relative to the discrepancy in the cooling power. This is likely due to the presence of bypass flow in the system (where some of the flow bypasses the beds and proceeds directly from the hot inlet valve to the hot outlet valve), which becomes more pronounced at flow rates above 16 liter min^{-1}. We note that when the flow rate between the cold outlet and cold inlet valves (which determines the cooling power) was 19.4 liter min^{-1}, the total flow rate produced by the pump was 20.3 Lit min-1, giving a bypass flow of 0.9 lit min^{-1}. The flow work associated with delivering this bypass flow appears in the measured heat exhaust, but is not included in the model. In contrast, the bypass flow at 12.5 and 15.5 liter min^{-1} was below the accuracy of the flow meters used in the system.

5. CONCLUSIONS

A large-scale magnetic refrigeration system, designed to meet the preliminary performance specifications for a naval supplemental electronics cooler, has been designed and constructed by

Astronautics. The system has produced over 3 kW of cooling power at zero span, and has achieved peak specific exergetic cooling power of 173 W tesla^{-1} liter^{-1}, corresponding to 2502 W over a span of 11.0 K with electrical COP of 1.9. Moreover, this COP was obtained using electrical components with mediocre efficiency. The system met its initial design goal by producing 2 kW of cooling power over a span of 12.0 K, although its energy efficiency was somewhat below the design goal. The cooling power and heat exhaust of the system were in generally good agreement with theoretical prediction.

ACKNOWLEDGEMENTS

Support for this effort was provided by the US Office of Naval Research under contracts N00014-10-C-0179 and N00014-11-C-0380.

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Figures and Tables

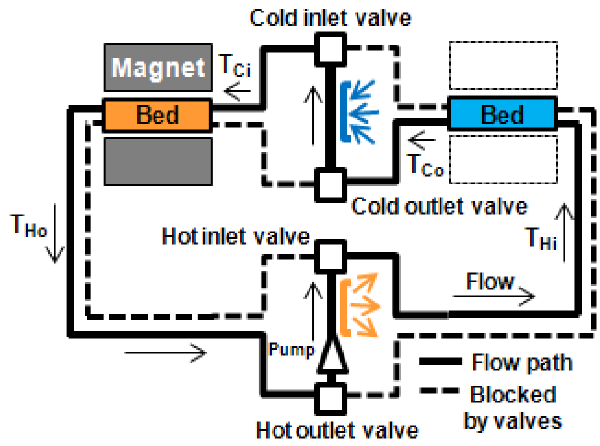

Figure 1. A schematic view of flow through a bed pair in the Astronautics system

Table 1. The Curie temperatures of the six layers of LaFeSiH used in the Astronautics system.

Layer	1	2	3	4	5	6
T_c (°C)	30.5	33.2	36.1	38.6	40.7	43.0

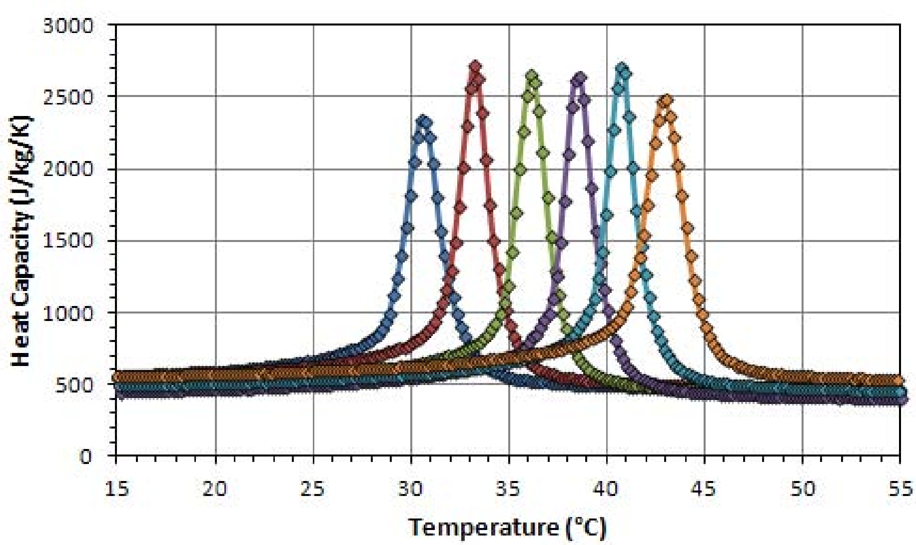

Figure 2. The zero-field heat capacities of the six layers of LaFeSiH (without epoxy) used in the Astronautics system.

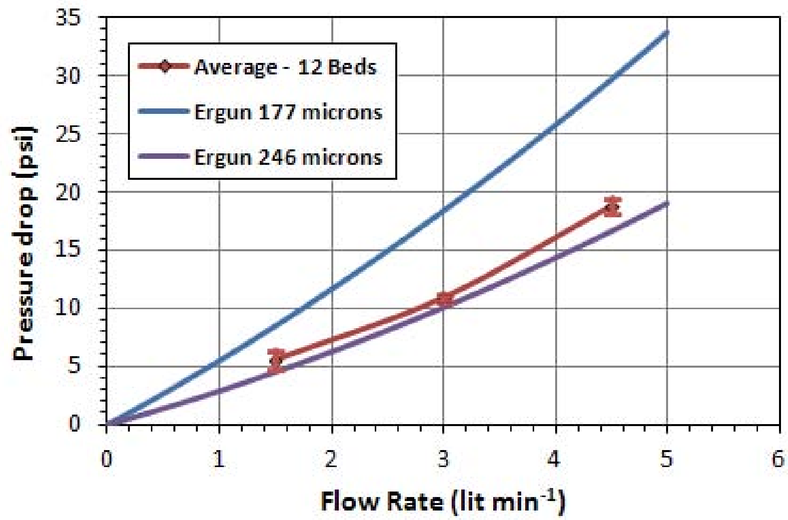

Figure 3. The average pressure drop of a bed used in the Astronautics system as a function of flow rate. The predictions of the Ergun-MacDonald equation are also shown for comparison.

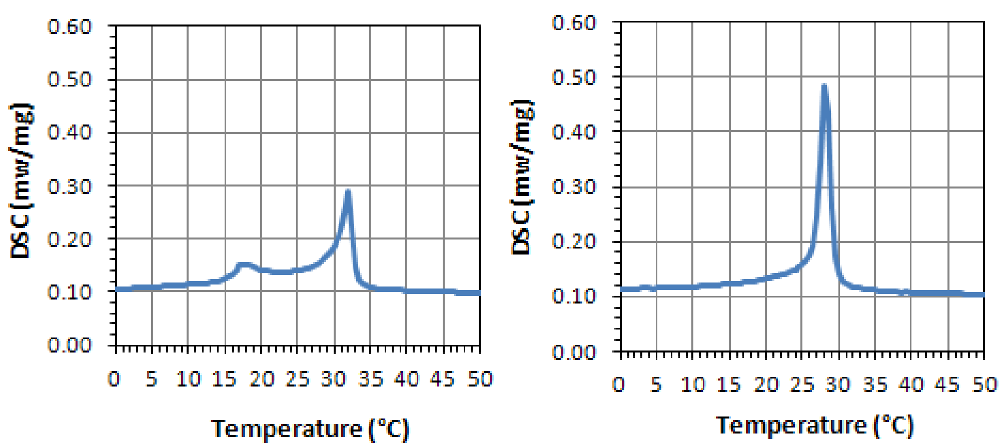

Figure 4. (Left panel) DSC trace of LaFeSiH after being held near its Curie temperature for over 1 year; (Right panel) Complete recovery of the material after 72-hour exposure at 60 °C.

Figure 5. The magnetic refrigeration system prior to (left) and after (right) insulation.

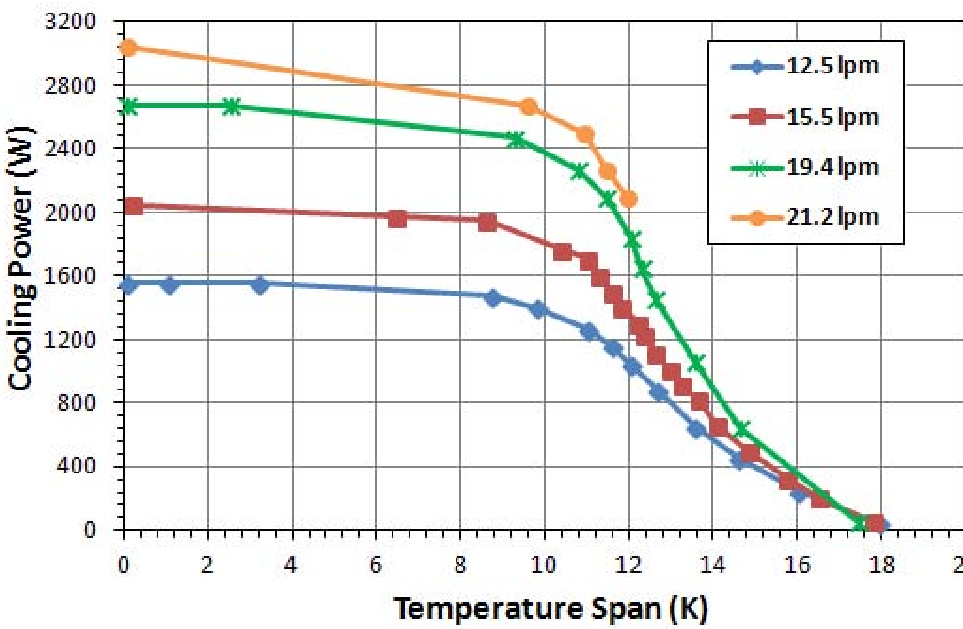

Figure 6. The cooling power of the system as a function of temperature span obtained at several different flow rates.

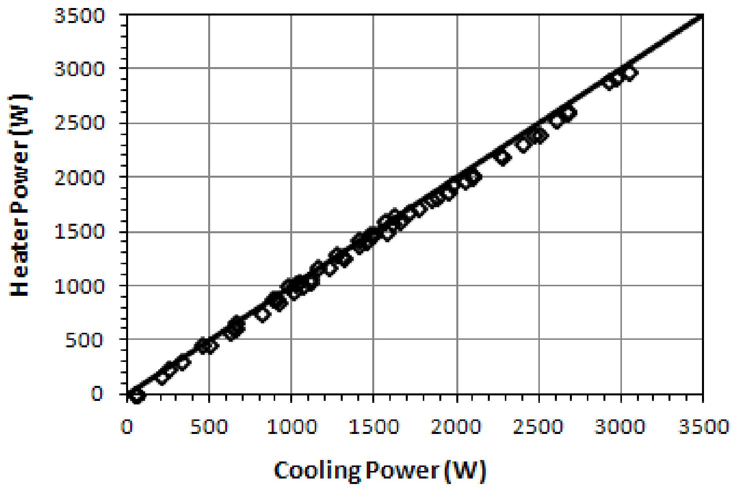

Figure 7. Comparison between heater power consumption and cooling power (1) of the system.

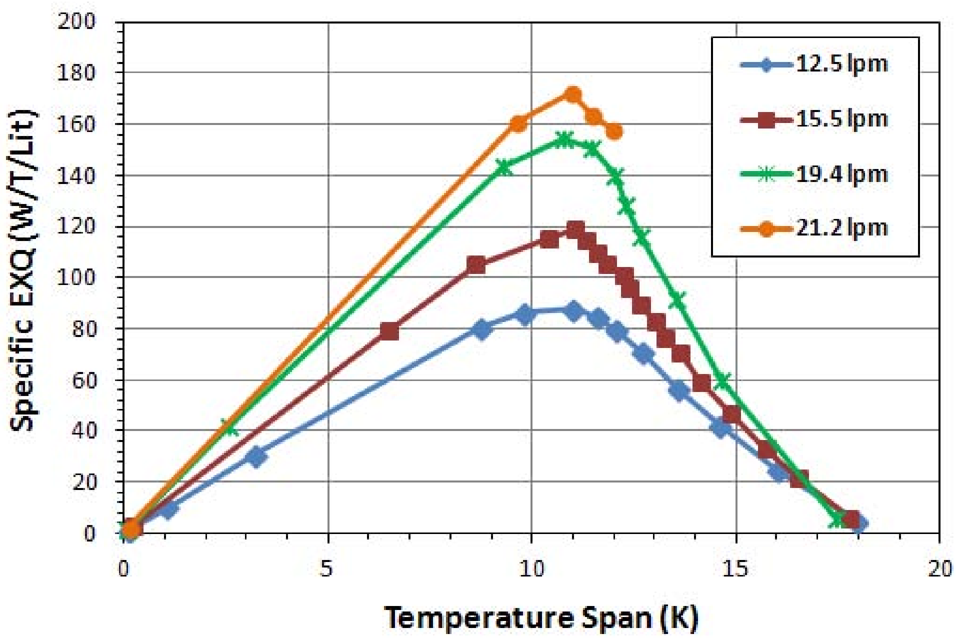

Figure 8. The specific exergetic cooling power (2) of the Astronautics system at various flow rates.

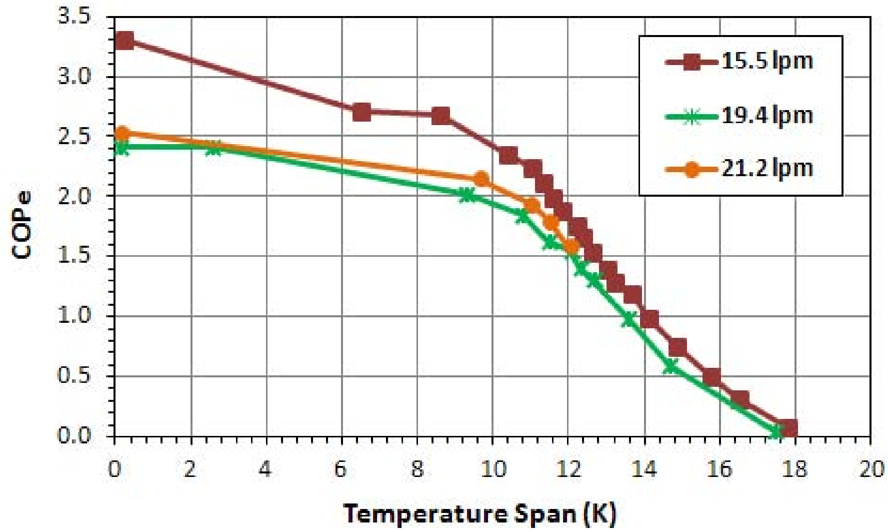

Figure 9. The electrical COP of the Astronautics system at flow rates 15.5, 19.4, and 21.2 liter min^{-1}.

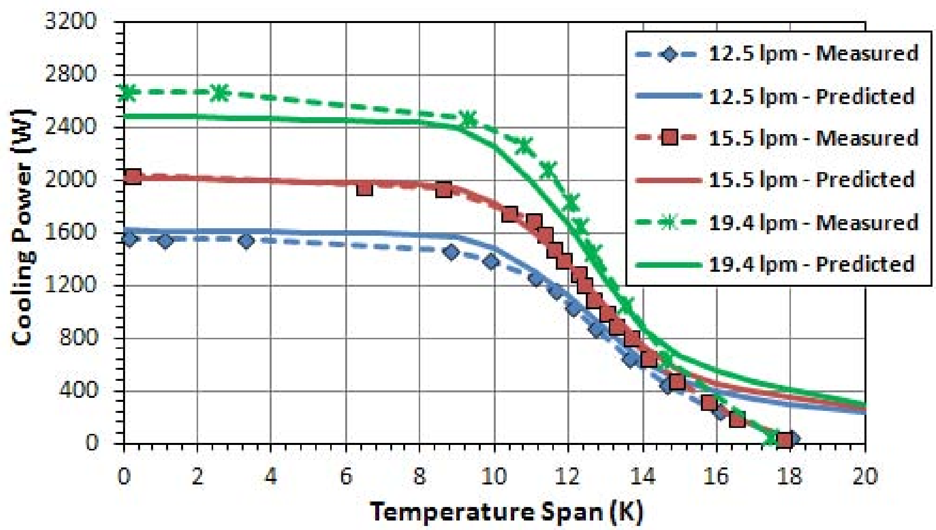

Figure 10. Measured and predicted cooling power versus temperature span of the Astronautics system.

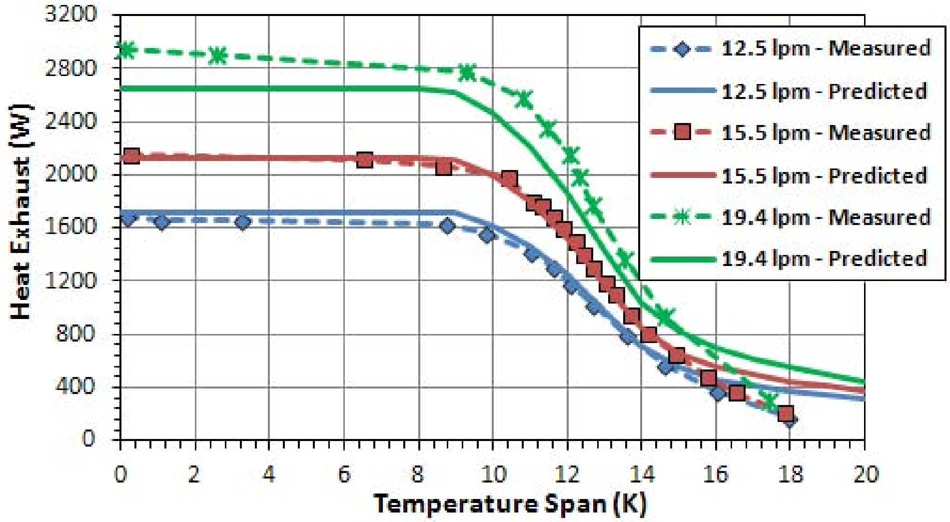

Figure 11. Measured and predicted heat exhaust versus temperature span of the Astronautics system.

Highlights for "**THE PERFORMANCE OF A LARGE-SCALE ROTARY MAGNETIC REFRIGERATOR"** by Steven Jacobs, Jon Auringer, Andre Boeder, Jeremy Chell, Leonard Komorowski, John Leonard, Steven Russek, and Carl Zimm

- Astronautics designed and constructed a new magnetic refrigeration system
- This system provided over 3 kW of zero-span cooling power
- The system provided 2.5 kW over a span of 11 °C
- These are the highest cooling powers yet reported for a magnetic system